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COURSE: ETHICAL HACKING

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Ethical Hacking – Protecting Systems and MFA

Definition

Ethical hacking is the process of legally testing computer systems, networks, or applications to find security weaknesses before malicious hackers can exploit them. Ethical hackers always have permission from the owner to perform these tests.

Uses of Ethical Hacking

1. Find Security Vulnerabilities – Detects weaknesses in websites, networks, and software.
2. Prevent Cyber Attacks – Helps stop hackers from stealing data or damaging systems.
3. Protect Personal and Business Data – Keeps sensitive information safe.
4. Improve System Security – Organizations fix vulnerabilities discovered during testing.
5. Meet Security Standards – Helps companies comply with cybersecurity regulations.
6. Build Customer Trust – Strong security increases confidence in online services.

Where Ethical Hacking is Used ?

- Banks and financial institutions
- Hospitals and healthcare systems
- Government organizations
- IT companies
- E-commerce websites
- Educational institutions
- Cloud computing platforms

Advantages

- Prevents data breaches
- Reduces financial losses
- Protects user privacy
- Improves cybersecurity awareness
- Strengthens overall network security

What is Multi-Factor Authentication (MFA)?

Multi-Factor Authentication (MFA) is a security method that requires two or more verification steps before allowing access to an account.

Common authentication factors:

1. Something you know - Password or PIN.
2. Something you have - Mobile phone or security token.
3. Something you are - Fingerprint or Face ID.

Example:

Digilocker it contain three step verification

Conclusion

Ethical hacking is a legal and responsible way to identify and fix security weaknesses before cybercriminals can exploit them. It plays a vital role in protecting digital systems and ensuring a safer online environment.